FitHub AI

“Train smarter. Live better.”

Comprehensive Software Project Life Cycle Document

Project Name	FitHub AI
Application Type	AI-Powered Fitness Coaching Web Application
Primary Tech Stack	Python (FastAPI), Jinja2, SQLAlchemy, SQLite, Google Gemini API, Docker, Uvicorn

2. Requirement Analysis Phase

Functional Requirements

- **Core Page Navigation:** Dedicated user interfaces for Home, Dashboard, Workout Planner, and Contact pages.
- **Workout Planner Inputs:** Interactive multi-field collection of parameters:
  - Goal (e.g., Lose fat)
  - Available time per session
  - Experience level & Training style
  - Days per week & Preferred days
  - Equipment available & Body focus
  - Recovery status
  - Body metrics: Height, Weight, Target Weight, and BMI goal
- **AI Engine & Output Rendering:**
  - System passes form parameters to Google Gemini API.
  - Displays dynamic workout card displaying estimated calorie burn, focus area, and difficulty level.
  - Generates categorized exercise breakdowns (Warm-up, Cardio, Upper Body, Legs, Core) with associated rep counts or durations.
- **Administrative Operations:** Dynamic configuration controls protected via an ADMIN_TOKEN environment variable.

Non-Functional Requirements

- **Performance:** High-concurrency, low-latency API handling powered by FastAPI and Uvicorn (ASGI).
- **Security & Environment Privacy:** Dynamic configuration loading from hidden .env files for key credentials (e.g., GOOGLE_API_KEY, ADMIN_TOKEN).
- **UI/UX:** Responsive, dark-themed interface with modern gradient accents and clean data typography.